

Inductor L2 and capacitors C7 and C13 form LC tank circuit tuned to the frequency of the LC



Fig. 4: Author's prototype showing the working with coils in perpendicular positions

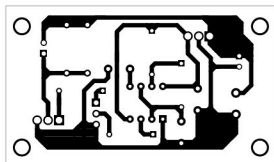


Fig. 5: Actual-size PCB pattern of the transmitter unit

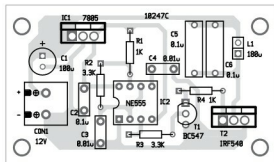


Fig. 6: Component layout of the PCB shown in Fig. 5

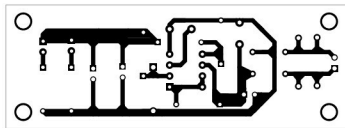


Fig. 7: Actual-size PCB pattern of the receiver unit

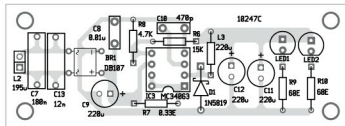


Fig. 8: Component layout of the PCB shown in Fig. 7

tank on transmitter side. The frequency of the receiver LC tank is 26kHz using the frequency formula explained earlier.

BR1 bridge rectifier is used to convert high-frequency AC to pulsating DC. C8 and C9 are smoothing capacitors for providing regulated DC voltage. To achieve high efficiency, buck/boost chip MC34063 is used. It has wide input voltage range of 3V - 40V. Its maximum output current is 1.5A with an adjustable output voltage.

In this circuit, it is configured as a buck converter or step-down converter. Output (V_{out}) of the buck converter which appears at C11 and C12 with respect to ground, is calculated as $V_{out} = 1.25V(1 + R6/R8)$ where, 1.25V is the internal reference voltage of MC34063 chip. So, $V_{out} = 1.25V(1 + 15k/4.7k) = 5.2V$, which is enough to drive power LEDs (LED1 and LED2).

R9 and R10 are the current-limiting resistors for LED1 and LED2.

Construction and testing

An actual-size, single-side PCB pattern for the transmitter unit is shown in Fig. 5 and its component layout in Fig. 6. Similarly, an actual-size, single-side PCB pattern for the receiver unit is shown in Fig. 7 and its component layout in Fig. 8. After assembling the circuits on the PCBs, enclose these in two separate boxes.

Load test. The receiver can deliver a load current

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V voltage regulator
IC2	- NE555 timer
IC3	- MC34063 buck/boost regulator
BR1	- DB107 bridge rectifier
T1	- BC547 npn transistor
T2	- IRF540 MOSFET
D1	- 1N5819 Schottky diode
LED1, LED2	- 1-watt LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R4	- 1-kilo-ohm
R2, R3	- 3.3-kilo-ohm
R5	- 150-ohm
R6	- 15-kilo-ohm
R7	- 0.33-ohm
R8	- 4.7-kilo-ohm
R9, R10	- 68-ohm

Capacitors:

C1	- 100 μ F, 25V electrolytic
R1, R4	- 0.1 μ F ceramic disk
C3, C4, C8	- 0.01 μ F ceramic disk
C5, C6	- 0.1 μ F, 100V polyester
C7	- 180nF, 100V polyester
C9, C11, C12	- 220 μ F, 25V electrolytic
C10	- 470pF, ceramic disk
C13	- 12nF, 100V polyester

Miscellaneous:

L1	- 180 μ H inductor using a 35-turn, 50mm diameter of 26SWG enamelled copper wire
L2	- 195 μ H inductor using a 45-turn, 50mm diameter of 26SWG enamelled copper wire
L3	- 220 μ H fixed inductor
	- 2-pin terminal connector
	- Heat sink for 7805

of 271mA at 5.2V constant voltage. Therefore approximate output power = $5.2V \times 271mA = 1.4W$, and power consumption of transmitter = $12V \times 180mA = 2.16W$ (approx.).

When the receiver is away from the transmitter, power consumed by transmitter = $12V \times 20mA = 240mW$.

Efficiency = output power/input power = $1.4/2.16 = 0.648$ or 64.8%

Because of the air-core coils (less magnetic coupling), efficiency is low. It can be improved by using magnetic-core based coils. **EFY**



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